

Seat No:

SET	P
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Pune University**Btech Endsem****Subject: Software Testing and Quality Assurance · Pattern: Section 1 24 Questions 70 Marks****Day & Date:** Wednesday, September 23, 2026 **Duration:** 180 Minutes **Max. Marks:** 34 Marks**Instructions for Candidates:**

1. Q.1 is compulsory. Mention the question paper set (**P**) clearly on top of the answer booklet.
2. Figures to the right indicate full marks assigned to each question.
3. Assume suitable data wherever necessary and state your assumptions explicitly.
4. Use of programmable calculators or unauthorized electronic communication devices is strictly prohibited.
5. 24 Marks Theory Based Questions

SECTION – I (Theory & Core Concepts)**[14 Marks]****Q.1 Answer the following questions (Attempt Any Four):****[16 Marks]**

- a) Explain why ISO-9001 standard and it's important in software testing? [6] [4]
- b) Examine the relationship between quality & productivity.[5] [5] [4]
- c) Define software quality. List & explain core component of quality.[5] OR [4]
- d) Illustrate selenium tool suite in detail.[6] [4]
- e) How would you explain selenium IDE explain in detail?[6] OR [4]

Q.2 Answer the following questions in detail (Attempt Any Two):**[12 Marks]**

- a) Write test cases for login validation.[5] documentation.[5] OR [4]
- b) Can you explain path coverage testing & conditional coverage testing. [6] [4]
- c) Compare the Ishikawa's Flow chart and Histogram tool.[6] OR [4]
- d) What do you think about static techniques?[6] OR [4]
- e) Explain the Six Sigma methodology and its characteristics in quality management. [6]
- f) Explain ISO 9001 and CMM/CMMI quality standards and their relevance to software testing. [6]
- g) Explain Statement, Branch, and Path coverage metrics with suitable code examples. [4]
- h) Explain the defect life cycle with a neat diagram. [4]

SECTION – II (Analysis, Design & Applications) [14 Marks]**Q.3 Answer the following questions (Attempt Any Four):** [16 Marks]

- a) Explain the concept of Regression Testing and when it should be performed. [3]
- b) Explain Decision Table based testing and Cause-Effect graphing techniques. [4]
- c) Explain Automation Testing frameworks (Data-Driven, Keyword-Driven, and Hybrid) with their benefits and limitations. [6]
- d) Explain Equivalence Class Partitioning and Boundary Value Analysis with examples. [4]
- e) Differentiate between Black-Box and White-Box testing techniques with examples. [6]

Q.4 Solve / Explain the following technical problems: [12 Marks]

- a) Compare Error Guessing and Exploratory Testing approaches. [4]
- b) Write a short note on Test Plan and Test Strategy documentation. [3]
- c) Explain Statement, Branch, and Path coverage metrics with suitable code examples. [4]
- d) Explain the defect life cycle with a neat diagram. [4]
- e) Compare Error Guessing and Exploratory Testing approaches. [4]
- f) Explain ISO 9001 and CMM/CMMI quality standards and their relevance to software testing. [6]
- g) Explain the Six Sigma methodology and its characteristics in quality management. [6]
- h) Explain Automation Testing frameworks (Data-Driven, Keyword-Driven, and Hybrid) with their benefits and limitations. [6]

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